

Yajvan Ravan

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EDUCATION

Massachusetts Institute of Technology (MIT)

June 2025

B.Sc. in Electrical Engineering & Computer Science; GPA: 5.0/5

- **Selected Courses:** Design & Analysis of Algorithms. Machine Learning. Computer Vision. Inference. Representation, Reasoning, & Inference in AI. Matrix Methods. Real Analysis. Linear Algebra. Sensorimotor Learning

EXPERIENCE

FortyFive Labs—Technical Staff

Jun 2025 - Dec 2025

- MIT CSAIL spin-off building open-source robotics tool suite and data hub
- Deployed Redis-based task queue into AWS and workstations allowing image generation at 50,000 images/hour
- Set up 20-node (5090 + A6000) in-house cluster with SLURM/Ansible for robot policy training/testing

Isola Lab, MIT—Research Intern (Advisor: Phillip Isola)

Oct 2024 - May 2025

- Built LucidXR, a VR data engine (>100k lines) to scale learning-from-demonstration for manipulation tasks
- Compiled MuJoCo sim to WebAssembly for in-browser physics simulation; developed visualization in ThreeJS.
- Implemented generative model policies (diffusion, VAE, transformer) with automated sim eval and video logging
- Trained visuomotor policies across 8 manipulation tasks (incl. deformable material, dexterous hands).
- Developed data pipeline converting raw MoCap poses to 10D action representations (Gram-Schmidt matrices).
- Developed keypoint trajectory augmentation (10x data amplification) and 3D Gaussian Splat evaluation suite
- Built distributed synthetic imagery generation pipeline with StableDiffusion + ControlNet + ComfyUI.
- **Achieved zero-shot sim-to-real transfer (zero-shot generalization) on pick-and-place.**
- **Co-first author publication in CoRL 2025.** <https://lucidxr.github.io/>

LIS Group, MIT—Robotics Research (Advisors: Leslie Kaelbling, Tomás Lozano-Pérez)

Aug 2023 - Sep 2024

- Developed PoPi, a hierarchical system for long-horizon mobile manipulation with unknown dynamics.
- Built data collection/processing system for Boston Dynamics Spot: synchronized RGB-D streams and proprioception, AprilTag-based localization, robot-to-world frame transforms
- Trained state-based, conditional UNet diffusion policy for short-horizon control within 2m.
- Implemented A*/RRT motion planners for SE(2) and offline mapping (clustering for obstacle detection).
- Designed hierarchical planner-policy architecture and full deployment stack; achieved 80% success on furniture rearrangement (vs. 0% pure learning, 50% pure planning). <https://yravan.github.io/plannerorderedpolicy/>.

NASA Langley Research Center—Research Intern

Jun 2023 - Dec 2023

- Built real-time wildfire detection system on multispectral satellite imagery (IR + thermal). Processed raw sensor data and trained UNet segmentation models. **First-author paper, AIAA SciTech 2026.**

AWARDS

Gold Medal at the 2019 International Chemistry Olympiad — 1st in the US, 19th in the world

2020 USA Math Olympiad Qualifier and **2019 USA Junior Math Olympiad Qualifier** — **Top 500** in US

PUBLICATIONS

Ravan, Y.*, Rashid, A.*, et al. Lucid-XR: An Extended-Reality Data Engine for Robotic Manipulation. *Conference on Robot Learning (CoRL) 2025*. <https://lucidxr.github.io/>.

Ravan, Y., et al. Combining Planning and Diffusion for Mobility with Unknown Dynamics. arXiv Preprint arXiv:2410.06911. <https://yravan.github.io/plannerorderedpolicy/>.

Yu, A.*, Yang, G.*, ... , **Ravan, Y.**, ... LucidSim: Learning Visual Parkour from Generated Images. *Conference on Robot Learning (CoRL) 2024*. <https://lucidsim.github.io/>.

Ravan, Y., et al. Real-Time Wildfire Localization on the NASA AMS using Deep Learning. *AIAA SciTech 2026*.

PROJECTS

MITScript Interpreter (C++): Interpreter with garbage collection, bytecode compiler, custom VM. Page

BeaverNav (Python): 10K+ line navigation app with pathfinding on 300+ floor plans at MIT. Page

SKILLS

Languages: C++, Python, C, JavaScript **Robotics:** Motion planning (A*, RRT, SE(2)), MuJoCo, Drake, IsaacLab, SolidWorks, Fusion 360, CAD Design, Fabrication, CNC **Systems:** Linux, Git, Docker, SSH, Web Assembly, SLURM **ML/Perception:** PyTorch, TensorFlow, OpenCV, diffusion models, Weights and Biases.